Independent Model Validation and Challenger Analysis

Credit Default Modeling with Logistic Regression, XGBoost, Calibration, Stability, and Model Risk Diagnostics

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Abstract

In this project, I develop an end-to-end independent model validation framework for a binary credit-default model. My objective is not merely to maximize predictive accuracy, but to determine whether a model is appropriate for its intended use, statistically reliable, sufficiently calibrated, stable under population change, interpretable, and governable. I begin with a structured data-quality and leakage review, build a transparent Logistic Regression model as the champion, and introduce XGBoost as a nonlinear challenger. I then evaluate discrimination, calibration, uncertainty, population stability, sensitivity, explainability, and post-implementation monitoring. My final objective is to support a model-risk decision of *Approve*, *Approve with Conditions*, or *Reject*. In my current repository, I have implemented the data-ingestion and independent data-audit layer; the remaining sections describe the validation methodology I plan to implement so that I can challenge and revise each modeling choice before fixing it in code.

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1 Project Objective

My objective in this project is to construct a model validation workflow that resembles the work of an independent Model Risk Management function. I deliberately approach the problem from the perspective of an independent validator rather than only from that of a model developer.

When I act as the model developer, I naturally ask how predictive performance can be improved. In this project, I then step into the role of an independent validator and ask a broader set of questions:

- Is the data fit for the intended modeling purpose?
- Is the methodology statistically justified and reproducible?
- Does performance generalize beyond the development sample?
- Are predicted probabilities well calibrated?
- Is the model stable under plausible population or feature shifts?
- Does a more complex challenger provide enough incremental value to justify additional model risk?
- Are material limitations explicitly documented and monitored?

I therefore organize my project around the validation chain

$$\text{Data} \rightarrow \text{Model} \rightarrow \text{Independent Testing} \rightarrow \text{Challenge} \rightarrow \text{Risk Decision}.$$

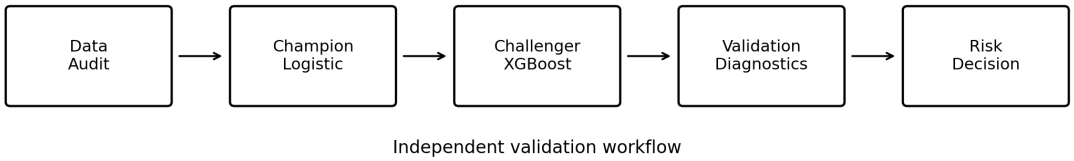


Figure 1: Independent model validation workflow I use throughout the project.

My guiding principle is to translate each validation result into four components:

$$\boxed{\text{Observation} \rightarrow \text{Risk} \rightarrow \text{Evidence} \rightarrow \text{Action}}.$$

I use this structure so that my work functions not only as a machine-learning exercise, but also as a model-risk case study.

2 Business Problem and Dataset

I model a binary credit-default indicator. For customer i, my model estimates

$$p_i = P(Y_i = 1 \mid X_i),$$

where $Y_i = 1$ denotes default and X_i contains observed customer and account characteristics.

I use the UCI *Default of Credit Card Clients* dataset in my starter repository. It contains 30,000 observations and 23 explanatory variables, together with a binary default outcome. The dataset is convenient for a validation project because it supports probability-of-default modeling, class-imbalance analysis, calibration assessment, and challenger-model comparisons.

In my current implementation, I retrieve the dataset through the `ucimlrepo` interface, separate predictors from the binary outcome, and keep the data-loading step reproducible inside the project source package.

Implemented in my current starter repository.

2.1 Important limitation: no genuine out-of-time sample

A genuine out-of-time design requires a clean timestamp that would let me train on earlier observations and evaluate on a later real-world period. In this project, the dataset does not provide a clean observation timestamp suitable for constructing that design.

This matters because a random holdout sample approximately answers the question

> Can the model generalize to unseen observations drawn from a broadly similar historical population?

whereas an out-of-time sample addresses the stronger question

> Will the model remain useful when the population and economic environment move forward in time?

The distinction is illustrated in Figure 2.

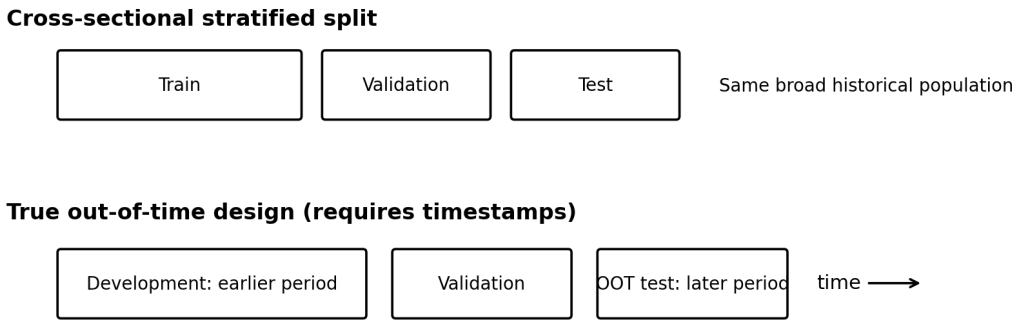


Figure 2: Cross-sectional holdout validation versus genuine out-of-time validation.

I therefore use a stratified train/validation/test split for the baseline analysis and later introduce controlled synthetic drift as a sensitivity exercise. I use synthetic drift to ask how the model reacts to a specified change; I do not treat it as a substitute for actual future-period evidence.

3 Independent Data Audit

In the first implemented stage, I ask whether I can trust the modeling data before fitting any model. This keeps me from jumping directly to model metrics while ignoring potentially material data issues.

3.1 Target prevalence

The event rate is

$$\widehat{\pi} = \frac{1}{n}\sum_{i=1}^{n} Y_i.$$

I first inspect the prevalence of defaults because raw classification accuracy can be misleading when the positive class is relatively uncommon. The default rate also affects the interpretation of precision, PR-AUC, Brier score, and calibration.

3.2 Missingness

For each variable X_j, the missing-value rate is

$$m_j = \frac{\#\{i : X_{ij} \text{ is missing}\}}{n}.$$

I do not want to impute missing values automatically. Important questions include whether missingness is informative, whether it is stable across samples, and whether the operational process generating missing values can change after deployment.

3.3 Duplicates

Duplicate observations can overweight particular records and can create train-test contamination if identical records appear in different partitions. The current audit reports both duplicate counts and duplicate rates.

3.4 Identifier and leakage risk

A near-unique variable may be an identifier rather than a meaningful predictor. The starter audit computes

$$\text{Unique Fraction}_j = \frac{\#\text{ unique values of } X_j}{n}$$

and flags unusually high-cardinality variables for review. The threshold is a diagnostic flag rather than an automatic deletion rule.

The most important leakage question is temporal: would the feature have been available at the actual decision time? A variable can be statistically predictive and still be invalid if it contains information observed only after the outcome window begins.

3.5 Audit deliverable

I will record each material finding in a table of the following form:

Table 1: Planned validation findings log.

ID	Area	Observation / Model Risk	Severity	Required Action
F-01	Data	Potential identifier or leakage field detected	–	Confirm timing and intended use
F-02	Data	Distribution or encoding anomaly	–	Investigate source and treatment
F-03	Sampling	Population representativeness concern	–	Document scope or revise sample

3.6 Conclusion of Independent Data Audit

Based on my initial data audit, I consider the dataset suitable for baseline model development, subject to several preprocessing and validation checks. Before fitting the final champion model, I will explicitly determine the treatment of numerically encoded categorical and ordinal variables, investigate unusual or undocumented category values, verify that no identifier-like or temporally unavailable variables enter the model, and document all transformations.

Because the dataset does not contain a reliable temporal structure, I will use a stratified train/validation/test split for baseline model development and treat the absence of genuine out-of-time validation as an explicit model-risk limitation. I therefore interpret the audit as a decision to proceed with modeling subject to these documented conditions, rather than as an unconditional approval of the data.

Implemented in my current starter repository.

4 Validation Design and Sample Splitting

My current splitter creates a 60/20/20 stratified train/validation/test design. Stratification approximately preserves the default rate in each subset. A fixed random seed is used so that the split can be reproduced exactly. I keep the splitting logic separately in the project source package rather than embedding it only inside a notebook.

The three samples have distinct purposes:

Table 2: Role of each modeling sample.

Sample	Approximate Share	Purpose
Training	60%	Fit preprocessing and model parameters
Validation	20%	Compare specifications and challengers
Test	20%	Final holdout after design choices are frozen

A key governance rule I follow is that preprocessing transformations must be learned from the training data rather than the full dataset. The final test set should not be repeatedly inspected during development, because repeated use turns it into another validation sample and weakens its independence.

5 Champion Model: Logistic Regression

I choose Logistic Regression as my champion because it provides a transparent probabilistic baseline and allows me to analyze coefficient direction, odds ratios, multicollinearity, calibration, and stability directly.

The model is

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

Equivalently,

$$p_i = \frac{1}{1+\exp(-\eta_i)}, \qquad \eta_i = \beta_0 + \beta^\top x_i.$$

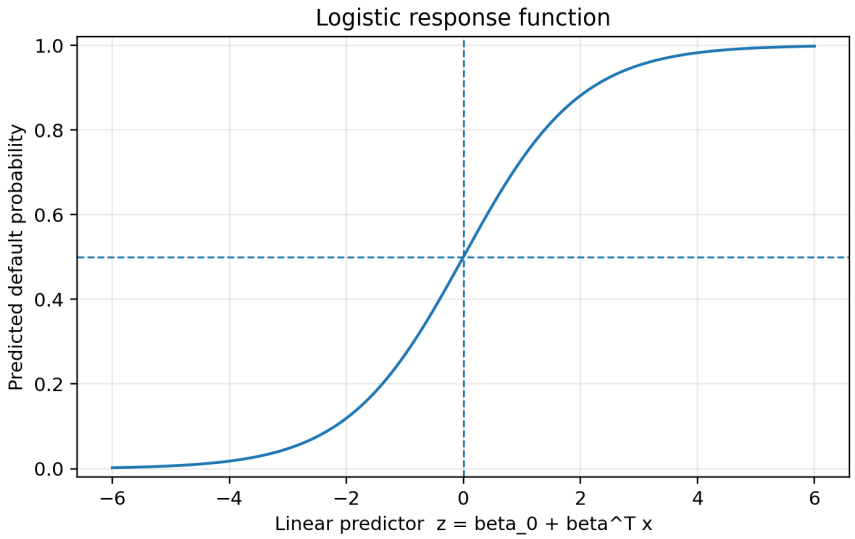


Figure 3: The logistic response function maps the linear predictor into a probability between zero and one.

For a one-unit increase in predictor x_j, holding the other predictors fixed, the odds are multiplied by

$$\exp(\beta_j).$$

My baseline implementation will use a reproducible preprocessing-and-model pipeline with scaling applied only from the training sample and an L_2-regularized Logistic Regression specification as the initial benchmark. Before I finalize this pipeline, I will explicitly decide how categorical and ordinal fields should be represented. Treating numerically encoded categories as continuous without justification can impose artificial linear structure.

Planned for the next implementation stage.

5.1 Champion-specific validation questions

I will examine:

- coefficient signs and economic or behavioral reasonableness,
- multicollinearity and coefficient instability,
- complete or quasi-separation,
- regularization choice,
- sensitivity to feature treatment,
- discrimination and calibration,
- segment-level stability.

A useful multicollinearity diagnostic is the variance inflation factor

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2},$$

where R_j^2 is obtained by regressing predictor X_j on the remaining predictors. High VIF does not automatically imply biased coefficients, but it can make individual coefficient estimates unstable and difficult to interpret.

6 Challenger Model: XGBoost

The challenger model is XGBoost. The purpose of the challenger is not to assume that a more complex model is superior. Instead, it tests whether nonlinear relationships and feature interactions contain materially useful predictive information that the champion misses.

I will begin the challenger with a deliberately conservative tree depth, learning rate, subsampling fraction, and feature-subsampling fraction. These settings are initial values rather than a final validated specification. I will deliberately constrain hyperparameter tuning so that I can distinguish genuine challenger value from validation-sample overfitting.

The challenger introduces additional risks:

- greater functional complexity,
- greater hyperparameter-selection risk,
- potentially weaker probability calibration,
- less transparent individual prediction logic,
- greater monitoring and explainability burden,
- potential sensitivity to population drift.

Therefore the relevant decision is not

Which model has the highest AUC?

but rather

Does the performance uplift justify the additional model risk?

7 Discrimination Analysis

Discrimination measures how effectively the model ranks defaulting observations above non-defaulting observations.

7.1 ROC-AUC

The area under the ROC curve can be interpreted approximately as

$$\text{AUC} = P(S^+ > S^-),$$

where S^+ is the score of a randomly chosen default observation and S^- is the score of a randomly chosen non-default observation.

AUC is useful because it is threshold-independent, but it does not tell us whether the predicted probabilities are numerically accurate.

7.2 Gini coefficient

For a binary model,

$$\text{Gini} = 2 \times \text{AUC} - 1.$$

7.3 Kolmogorov–Smirnov statistic

A common credit-model diagnostic is

$$\text{KS} = \sup_t |F_0(t) - F_1(t)|,$$

where F_0 and F_1 are the empirical score distributions for non-default and default observations.

7.4 Precision-recall behavior

When the positive class is relatively uncommon, precision-recall behavior provides an important complement to ROC-AUC. I will therefore report PR-AUC together with threshold-level precision and recall.

Table 3: Planned discrimination output.

Metric	Logistic Champion	XGBoost Challenger
ROC-AUC	To be estimated	To be estimated
PR-AUC	To be estimated	To be estimated
Gini	To be estimated	To be estimated
KS	To be estimated	To be estimated

8 Probability Calibration

A credit-risk model produces probabilities, not only rankings. Calibration therefore receives separate treatment from discrimination.

Suppose a set of customers receives predicted default probabilities around 20%. If the model is well calibrated, approximately 20% of comparable observations should default over the relevant outcome horizon.

8.1 Reliability diagram

Predictions are grouped into probability bins. For bin b, I compare

$$\overline{p}_b = \frac{1}{n_b} \sum_{i \in b} \widehat{p}_i$$

with the realized event rate

$$\overline{y}_b = \frac{1}{n_b} \sum_{i \in b} y_i.$$

Perfect calibration lies near the 45-degree line.

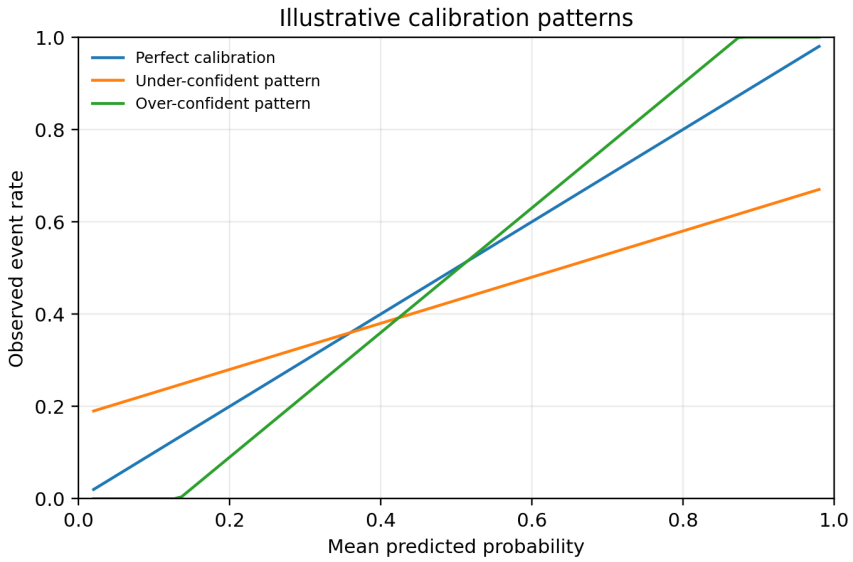


Figure 4: Illustrative calibration patterns. This figure is conceptual and does not represent my current empirical results.

8.2 Brier score

The Brier score is

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^{n} (\widehat{p}_i - y_i)^2.$$

Lower values indicate smaller squared probability errors, although interpretation should be made relative to event prevalence and appropriate benchmarks.

8.3 Calibration intercept and slope

I will also estimate

$$\operatorname{logit} P(Y = 1) = a + b \operatorname{logit}(\widehat{p}).$$

The ideal values are approximately $a = 0$ and $b = 1$. A slope below one often indicates predictions that are too extreme, while an intercept shift can indicate systematic over- or under-prediction.

9 Uncertainty and Bootstrap Validation

A single metric value can create a false impression of precision. I will use bootstrap resampling to quantify uncertainty around selected validation metrics and challenger improvements.

For bootstrap iteration b:

1. sample n observations with replacement from the validation sample,
2. recompute the metric,
3. repeat for B iterations,
4. summarize the empirical bootstrap distribution.

I will resample validation observations with replacement, recompute the selected metric in each replicate, and obtain empirical confidence intervals from the resulting distribution. For champion-challenger comparison, a paired bootstrap will be preferable because both models are evaluated on the same resampled observations. The distribution of

$$\Delta_b = \text{Metric}^{(b)}_{\text{challenger}} - \text{Metric}^{(b)}_{\text{champion}}$$

will show whether apparent improvement is stable rather than driven by sampling variation.

Planned for the next implementation stage.

10 Population Stability and Drift

Model performance can deteriorate because the population changes after development. Two broad changes are important:

$$P_{\text{new}}(X) \neq P_{\text{development}}(X)$$

and

$$P_{\text{new}}(Y \mid X) \neq P_{\text{development}}(Y \mid X).$$

The first is a change in the covariate distribution; the second is a change in the relationship between predictors and outcome.

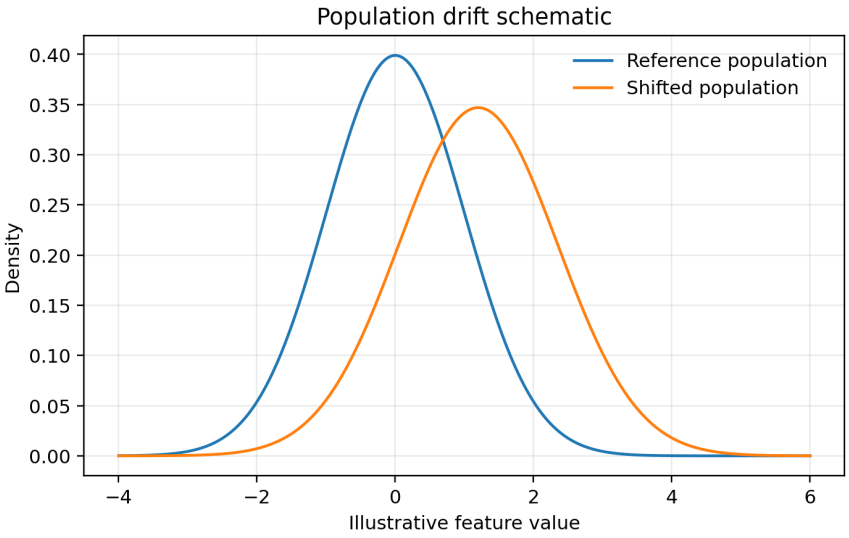


Figure 5: Illustrative population drift. The shifted distribution represents a hypothetical future population rather than an empirical project result.

10.1 Population Stability Index

For bins $i = 1, \ldots, B$, PSI is computed as

$$\mathrm{PSI} = \sum_{i=1}^{B} (A_i - E_i) \log\left(\frac{A_i}{E_i}\right),$$

where E_i is the reference population share and A_i is the comparison population share.

PSI is treated as a diagnostic rather than a universal pass/fail rule. Any practical threshold should be justified for the specific model and monitoring context.

Because my current dataset lacks a genuine future-period sample, I will distinguish clearly between actual observed drift and synthetic stress scenarios.

11 Sensitivity and Stress Testing

Sensitivity testing asks whether model behavior is reasonable when inputs are perturbed. The objective is not to predict the exact future, but to identify fragile or implausible behavior.

Example tests include:

- increasing utilization or balance variables,
- worsening payment-history indicators,
- shifting the distribution of credit limits,
- introducing additional missingness,
- perturbing important continuous features by $\pm 5\%$ or $\pm 10\%$.

For a continuous feature x_j, a local sensitivity measure can be approximated by

$$S_j \approx \frac{\widehat{p}(x_j + \delta) - \widehat{p}(x_j)}{\delta}.$$

I will focus on unexpected directions, discontinuities, excessive concentration of sensitivity in one feature, and large changes caused by small operational perturbations.

12 Explainability and Behavioral Reasonableness

Explainability is model-specific.

12.1 Logistic Regression

For the champion, I will examine coefficient signs, odds ratios, coefficient stability, and selected confidence intervals or resampling distributions where appropriate.

A positive coefficient implies an increase in log-odds as the predictor increases, holding other included variables fixed. However, I will avoid interpreting these coefficients causally unless the identification assumptions required for causal inference are separately justified.

12.2 XGBoost

For the challenger, I will use global feature importance and SHAP-based summaries. Local explanations may also be examined for representative observations.

An important validation principle is that explanation tools describe the fitted model; they do not prove that the model is unbiased, stable, causal, or appropriate. Therefore, I will use SHAP analysis as one diagnostic among many rather than as a substitute for validation.

13 Champion–Challenger Decision Framework

I will combine several dimensions in my final model comparison.

Table 4: Planned champion–challenger decision dimensions.

Dimension	Logistic Champion	XGBoost Challenger	Validation Question
Discrimination	Baseline	Potential uplift	Is improvement material and stable?
Calibration	Usually transparent	May require recalibration	Which probabilities are more reliable?
Interpretability	High	Lower	What explanation burden does the use case require?
Stability	Easier to diagnose	Potentially more fragile	Does performance persist under shift?
Complexity	Lower	Higher	Is complexity justified by incremental value?
Monitoring	Moderate	Higher	Can ongoing risk be monitored effectively?

The preferred model is therefore defined conceptually as

$$\boxed{\text{Best risk-adjusted evidence, not highest single metric}}.$$

A challenger could have a slightly higher AUC yet still fail to replace the champion if the uplift is uncertain, calibration is materially weaker, or stability deteriorates under plausible shifts.

14 Model Risk Discussion

I summarize the main model-risk issues I expect to investigate below.

Table 5: Key model-risk themes in the planned validation.

Risk Area	Real-World Issue	Validation Response
Data quality	Missingness, coding anomalies, identifiers	Independent audit and findings log
Leakage	Information unavailable at decision time	Feature timing review
Sampling	Cross-sectional holdout may not represent future periods	Explicit limitation and stress tests
Class imbalance	Accuracy can be misleading	PR metrics and threshold analysis
Calibration	Strong ranking can coexist with poor probabilities	Reliability curve and Brier score
Overfitting	Challenger can exploit validation noise	Conservative tuning and bootstrap comparison
Drift	Population can change after development	PSI, distribution monitoring, stress tests
Explainability	Complex model may be harder to challenge	SHAP plus behavioral reasonableness checks
Implementation	Training logic may differ from production logic	Reproducible pipeline and implementation checks

From a model validation perspective, the most important questions are:

- Is the data representative of the intended population?
- Are all model inputs available at decision time?
- Is Logistic Regression sufficiently flexible for the problem?
- Does XGBoost provide a material and stable improvement?
- Are predicted default probabilities calibrated?
- How sensitive are predictions to plausible population shifts?
- What limitations should I disclose before model use?

15 Post-Implementation Monitoring

Approval is not the end of validation. A production model requires monitoring capable of detecting deterioration.

Table 6: Proposed monitoring framework.

Monitoring Item	Example Signal	Possible Action
AUC / KS	Material decline from validation benchmark	Investigate performance deterioration
Calibration	Systematic over- or under-prediction	Recalibrate or redevelop
Feature PSI	Persistent population shift	Assess representativeness and sensitivity
Missingness	Sudden increase in operational fields	Investigate data pipeline
Score distribution	Unexpected compression or expansion	Review drift and implementation
Segment performance	Deterioration concentrated in subgroup	Restrict use or remediate

I will not treat monitoring thresholds as universal constants. I will establish them using model materiality, historical variability, business use, and risk appetite.

16 Validation Recommendation Framework

I will conclude the final report with one of three broad outcomes:

Table 7: Model-risk recommendation categories.

Decision	Interpretation
Approve	Evidence is strong and limitations are adequately controlled.
Approve with Conditions	Model is usable, but remediation, recalibration, enhanced monitoring, or documentation is required.
Reject	Material data, methodology, implementation, calibration, stability, or governance weaknesses remain unresolved.

My completed conclusion will contain the intended use, material strengths, validation findings, challenger evidence, residual limitations, required remediation, monitoring expectations, and final decision.

An example of the style of conclusion I may eventually reach is:

> The challenger improves discrimination, but the uplift is modest relative to estimation uncertainty and is accompanied by weaker calibration and greater sensitivity to population shift. Retain the Logistic Regression champion subject to recalibration and enhanced drift monitoring.

This is only an example of the type of model-risk reasoning I want to demonstrate; it is not one of my current empirical results.

17 Current Implementation Status

In my current starter repository, I have implemented the following components:

- UCI dataset loading,
- data-quality summary,
- duplicate summary,
- target prevalence summary,
- suspicious high-cardinality flags,
- 60/20/20 stratified sample splitting,
- initial unit tests for the audit functions.

I have deliberately left the following components for later implementation so that I can review the statistical design first:

- categorical-variable treatment,
- Logistic Regression champion,
- multicollinearity diagnostics,
- XGBoost challenger,
- discrimination and calibration reports,
- bootstrap uncertainty,
- PSI and drift analysis,
- sensitivity and stress testing,
- SHAP and explainability,
- final findings log and validation recommendation.

This separation is intentional. I want to understand and justify the methodology before converting the report into a fixed implementation.

18 Implementation Details

I implement the project in Python. The main packages are pandas and NumPy for data manipulation, scikit-learn for modeling and metrics, statsmodels for statistical diagnostics where useful, XGBoost for the challenger model, SHAP for explanation, and matplotlib for visualization.

I organize the implementation into modular components so that I can test analytical logic independently from presentation notebooks.

Table 8: Planned implementation components.

Component	Python Implementation / Responsibility
Data ingestion	`data.py`: load and standardize the source dataset
Independent audit	`audit.py`: missingness, duplicates, cardinality, target summaries
Sample splitting	`split.py`: reproducible stratified partitions
Preprocessing	`preprocessing.py`: encoding, scaling, transformations
Champion and challenger	`models.py`: Logistic Regression and XGBoost
Performance metrics	`metrics.py`: ROC-AUC, PR-AUC, KS, Gini, thresholds
Calibration	`calibration.py`: reliability curve, Brier score, slope/intercept
Uncertainty	`bootstrap.py`: resampling-based confidence intervals
Population stability	`stability.py`: PSI and distribution-shift diagnostics
Sensitivity testing	`stress.py`: controlled feature and population perturbations
Explainability	`explainability.py`: coefficients, odds ratios, SHAP summaries

I will keep reusable logic in the `src/` package and use notebooks for analysis, interpretation, and reporting rather than placing all modeling logic inside notebook cells. I maintain the detailed code examples separately in my project code companion.

19 Conclusion

I designed this project to demonstrate my complete thought process for independent model validation rather than only model construction. I begin with data quality and leakage review, establish a transparent Logistic Regression champion, challenge it with XGBoost, and then evaluate both models using discrimination, calibration, uncertainty, stability, stress testing, and explainability diagnostics.

The most important conceptual distinction is that predictive performance alone is not sufficient evidence of model validity. A model can rank observations effectively while producing poor probabilities; it can perform well in a random holdout yet deteriorate under future population change; and a complex challenger can improve a metric while introducing additional instability, governance burden, or calibration risk.

I therefore treat the final model decision as a risk-adjusted judgment. My ultimate deliverable is not simply a performance table, but a validation conclusion supported by evidence, documented limitations, and monitoring requirements.

At the current stage, I have implemented the data-audit and splitting infrastructure while intentionally specifying the remaining analytical modules before coding them. This lets me modify the methodology as the statistical assumptions, data characteristics, and validation findings become clearer.

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